Exam Preparation Notes

Quick revision for Python, Data Science, and Machine Learning

1. Python Quick Revision

- input() reads text; convert using int() or float() for numeric operations.
- Use == for equality and = for assignment.
- Lists are mutable; tuples are immutable; sets store unique values; dictionaries store key-value pairs.
- range(1, n + 1) includes values from 1 through n.
- A function uses def and can return a value using return.
- Use try/except to handle expected runtime errors.

```
n = int(input())
if n % 2 == 0:
print("Even")
else:
print("Odd")

total = 0
for i in range(1, n + 1):
total += i
print(total)
```

2. Data Science Quick Revision

- Workflow: define problem, collect, clean, explore, analyse/model, communicate, monitor.
- CSV stores tabular text; JSON stores key-value data and can be nested.
- Common data-quality issues include missing values, duplicates, inconsistent labels, and incorrect types.
- Bar charts compare categories; histograms show distributions; line charts show trends; scatter plots show relationships.
- Correlation does not prove causation.
- The median is often more robust to outliers than the mean.

```
import pandas as pd
df = pd.read_csv("data.csv")
print(df.head())
print(df.shape)
print(df.info())
print(df.isna().sum())
print(df.describe())
```

3. Machine Learning Quick Revision

- Supervised learning uses labelled data; unsupervised learning finds patterns without target labels.
- Regression predicts numbers; classification predicts categories; clustering groups observations.
- Training data fits the model; validation data supports model selection; test data estimates final performance.
- Overfitting means the model learns training noise and generalizes poorly.
- Precision focuses on correctness of positive predictions; recall focuses on finding actual positives.
- Data leakage can create unrealistically optimistic evaluation results.

4. Formula Sheet

Basic statistics

- Mean = sum of observations / number of observations.

- Range = maximum – minimum.
- Standard deviation = square root of variance.

Classification

- Accuracy = (TP + TN) / (TP + TN + FP + FN).
- Precision = TP / (TP + FP).
- Recall = TP / (TP + FN).
- F1 = 2 × Precision × Recall / (Precision + Recall), when the denominator is non-zero.

Regression

- MAE = average absolute error.
- MSE = average squared error.
- RMSE = square root of MSE.

5. Practice Questions with Short Answers

1. What is Data Science? — Using data, statistics, programming, and domain knowledge to generate insights.
2. Why clean data? — To improve reliability and reduce errors.
3. Which chart compares categories? — Bar chart.
4. Which chart shows a numerical distribution? — Histogram.
5. What is supervised learning? — Learning from examples with known target labels.
6. What is overfitting? — Learning training noise that does not generalize.
7. Why split data? — To evaluate performance on unseen examples.
8. Does correlation prove causation? — No.
9. Which Python collection stores key-value pairs? — Dictionary.
10. What does df.head() do? — Shows the first five rows by default.

6. Last-Day Study Plan

1. Review key definitions and the Data Science workflow.
2. Practise Python variables, conditions, loops, collections, functions, and files.
3. Revise statistics and match chart types to questions.
4. Review ML learning types, common algorithms, and evaluation metrics.
5. Answer practice questions without looking at notes, then correct mistakes.

These are general revision notes, not an official university question paper or syllabus. Compare them with your current course outline and faculty instructions.